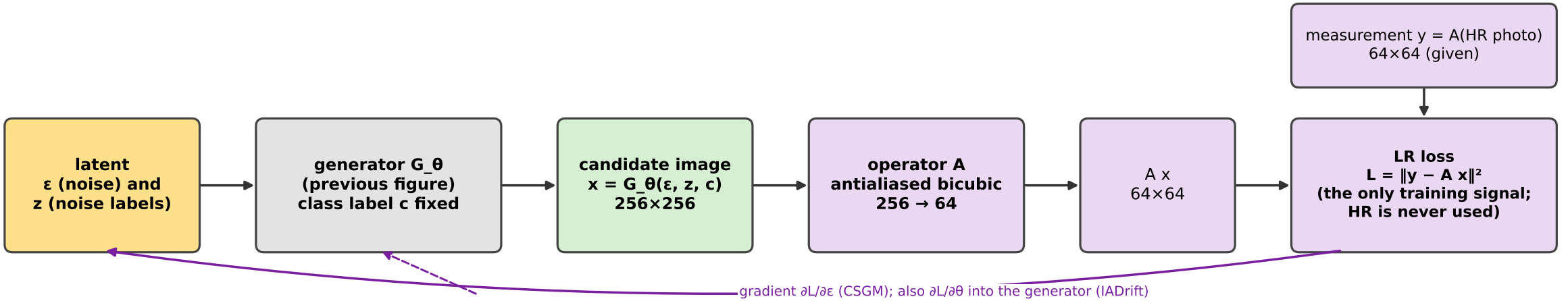
What we did: super-resolution ×4 by inverting the frozen / adapted drift generator



Method	What is optimized / done (generator, A, y as above)	Outcome (evidence)
CSGM	optimize ε (Adam, 5,000 steps, 2 restarts); z random & fixed; θ frozen	*test: + BP 22.75 dB / LPIPS 0.284 (bicubic + BP: 25.57 dB / 0.358)*
IADrift (IAGAN-style)	start from CSGM; optimize θ (all 134M weights) and ε jointly, 100 / 1,000 steps	*test, + BP vs CSGM + BP: @100 −0.12 dB / −0.008 LPIPS; @1000 −2.92 dB / +0.082*
noise-label search	try several random z; keep the one with the lowest LR loss (z is discrete)	*dev only (4 images): small gain (+0.24 dB / −0.013 LPIPS after BP)*
soft z / z swaps	relax z to embedding mixtures, or swap single z_i during CSGM	*dev only: no gain after BP*
back-projection (BP)	$x \leftarrow x + A^T(AA^T)^{-1}(y - A\,x)$: closest image that reproduces y exactly	*test: helps every method, 100/100 images (CSGM +1.05 dB, LPIPS −0.080)*
ensemble (posterior mean)	average 5 CSGM runs (2 restarts + 3 on shifted y), weighted by LR loss, then BP	*test (post-hoc): +1.57 dB, 100/100 images; LPIPS −0.032 vs CSGM + BP*

Yellow = latent / weights being optimized or selected; purple = measurement side. Main finding: the generator can represent the photos (fit to HR: 40 dB), but SR leaves everything above the LR limit unconstrained; more freedom (θ) adds wrong high-frequency detail, while averaging several reconstructions + BP removes it. test = 100 held-out images (post-hoc: added after the pre-registered analysis); dev = development images.